

Describing the Cleaned NHANES Dataset

Charles-Antoine Auger

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1 Introduction

The National Health and Nutrition Examination Survey (NHANES) is a continuous programme of the National Center for Health Statistics that combines interviews, physical examinations and laboratory measurements to describe the health and nutritional status of the civilian, non-institutionalised population of the United States (National Center for Health Statistics 2020). Participants are selected through a complex multistage probability design, and data are released in two-year waves.

This report describes the cleaned NHANES extract used throughout BIOS 640. The extract contains 29,400 participants across three waves and retains demographic variables, four repeated systolic and diastolic blood pressure readings, and a small set of socioeconomic indicators. Two tables summarise the dataset: the first covers sample composition, the second the distribution of blood pressure. All analyses were carried out in R (R Core Team 2025), and the tables were formatted with kableExtra (Zhu 2024).

2 Sample composition

Table 1: Composition of the cleaned NHANES sample by survey wave

	NHANES wave		
	2015	2016	2017
Sample size			
Participants	10,175	9,971	9,254
Sex, n (%)			
Male	5,003 (49.2)	4,892 (49.1)	4,557 (49.2)
Race and Hispanic origin, n (%)			
White	3,674 (36.1)	3,066 (30.7)	3,150 (34.0)
Black	2,267 (22.3)	2,129 (21.4)	2,115 (22.9)
Mexican	1,730 (17.0)	1,921 (19.3)	1,367 (14.8)
Asian	1,074 (10.6)	1,042 (10.5)	1,168 (12.6)
Other Hispanic	960 (9.4)	1,308 (13.1)	820 (8.9)
Other	470 (4.6)	505 (5.1)	634 (6.9)

Note:

Percentages are column percentages, computed within each wave. Sex reflects the original NHANES variable, which records biological sex rather than gender identity.

The three waves are of comparable size, ranging from 9,254 to 10,175 participants. The sex distribution is remarkably stable: males represent between 49.1% and 49.2% of each wave, so any difference observed across waves cannot be attributed to a shift in sex composition.

Race and Hispanic origin vary more. The White group is the largest in every wave, but its share moves by more than five percentage points, from 36.1% in 2015 to 30.7% in 2016. Other Hispanic participants show the opposite pattern, peaking at 13.1% in 2016 before falling back to 8.9% in 2017. These movements appear to reflect the oversampling strategy NHANES applies to selected subgroups, which changes between survey cycles, rather than a genuine change in the underlying population.

3 Blood pressure

Table 2: Average systolic blood pressure by survey wave, among participants with at least one valid reading

Wave	n	Average systolic blood pressure (mm Hg)				
		Min	Median	Mean	SD	Max
2015	7,528	64.7	115.3	118.3	18.1	228.7
2016	7,363	74.0	117.3	120.4	18.4	231.3
2017	6,713	72.7	118.0	121.7	20.3	234.0

Note:

The average systolic blood pressure is the mean of up to four readings taken at a single examination, and is missing only when all four readings are unavailable. Roughly one participant in four has no valid reading (26.0%, 26.2% and 27.5% across the three waves). The mean is shown in bold to highlight the upward trend.

Average systolic blood pressure rises steadily across the three waves, from 118.3 mm Hg in 2015 to 120.4 in 2016 and 121.7 in 2017, a total increase of 3.4 mm Hg. The medians move in the same direction, which suggests that the shift affects the bulk of the distribution rather than being driven by extreme values alone.

The spread also widens, with the standard deviation increasing from 18.1 to 20.3 mm Hg. The maximum values sit close to 230 mm Hg in every wave, so the upper tail is not what changed; rather, more participants fall further from the centre in the later wave. Because the waves differ in their race and Hispanic origin composition, and because age is a strong determinant of blood pressure, this trend should be read as a description of these samples rather than as evidence of a change in the underlying population.

References

- National Center for Health Statistics. 2020. *National Health and Nutrition Examination Survey Data*. Centers for Disease Control; Prevention. <https://www.cdc.gov/nchs/nhanes/>.
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